

HIROSHIMA UNIVERSITY

TRANSDISCIPLINARY IMMERSION PROGRAM

- Research $\Rightarrow$ Implementation ^{of research} into society is prioritized.
- Campus 4D $\Rightarrow$ Research/project is 1st implemented in campus before society.

2nd Period Intro to HU Program
Mr. Hatakeyama

- Applying for HU

$\hookrightarrow$ IAAS (International Admissions Assistant System)

$\hookrightarrow$ Financial support.

$\hookrightarrow$ Govt. (MEXT) program. (13%)

(946 USD pm)

$\hookrightarrow$ Private Scholarships (23%)

$\hookrightarrow$ HU scholarship (X)

NOTE $\rightarrow$ No scholarships for 3+1

Date: 30/1/25

3rd Session

Mr. Sunakawa

Intro to 3+1 program.

- Started in 2024 (English) 2016 (Japanese)
- Spring, fall semester. (2)
- 9 months (3 terms)
- 399600 yen
- NO DEGREE from HU (only certificate)

31/1/25

DAY-2

Session 1

→ Takao Momo

Fundamentals of Japanese Anime

- ↳ one of the six major industry
- ↳ 3 Trillion Yen industry
- ↳ 2004 Govt. law.
- ↳ Studios (811) → Mostly in Tokyo
- ↳ Two types of Business Models.

- 13 phases Theory of Kanekoisism
Takao

Session 2

Dr. Vinayak Bhanage

AI in Geospatial Technology. (bhanage@hiroshima-u.ac.jp)

↳ Intro to Artificial Intelligence

↳ Components of AI.

↳ Geospatial Technology Spectrum

↳ GNSS & positioning

↳ Earth observation

↳ GIS/spatial Analytics.

↳ 3D scanning.

↳ GeoAI is needed. for various reasons (DL, save time, analysis, accurate models)

↳ Accuracy already exists for models, the new task at hand is to optimize the computational resources (reduce cost and time)

↳ Currently {Neumrical methods} are being used.

(AIR QUALITY)

↳ Sentinel 5p → European Space agency. (provides air quality data)

↳ Fog estimation using AI.

↳ Pollution Trajectory (NASA)

Date: 31/1/25

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Session 3

Sushil Raut

High speed camera, robotics, mechatronics.

3/2/25

Session 1

India - Japan S\$T cooperation

by Dr. Yashwant Dev Parwar

(Indian Embassy)

→ Sakura Science program.
(India specific - Internships)

→ India has many programs.

→ Workshops, projects proposals, etc.

Session 2

TCS Automotive Engineering

by Amol Waghmare.

→ 400 employees every year.

→ Bilingual & certain software.

→ catia, Krio, unographics, abacus, hypermesh, ... → (softwares)

→ easy to apply in Japan TCS if you know the language.

→ In India, 2 exams to join TCS.

(basic + advanced)

Session 3 Advanced Research in Transport

Engineering / planning (CAI)

by Makoto Chikaraishi

- AI → Many applications but not yet confirmed if it good for society
- Ethical dilemma of self-driving cars
- MATSim → Transportation simulator

Session 4 Optimal Networks

→ Victor Parque

- Genetic Network Programming (GNP)
- Convolution Neural Networks
- Solve the problem faster than (g), you can write a paper on it.
- Minimal Trees

Date: 5/1/25

Session 1

AI

→ Deepak Kumar

- SWOT Analysis of life in Japan.
- How to crack interview in Japan.
- Sensyn core (Drone images with infrastructure (AI))
- Indo Banaji Scholarship, MEST, Jaso

Session 2

Campus to Career {Semiconductor Industry}

→ Atul Sharma

- Synaptics Japan (company)
- Core technical skills for RTL engineers.
- ASIC Designing & fabrication process.
- Leveraging Ph.D for RTL design career.
- Local Dimming IP design process.
- Foveal IP
- Display driver ICs.

Session 3

→ Combustion & Explosion

→ Wookyoung Kim.

- understanding explosion from industry to space
- what is combustion.

	Gasoline	Kerosene
Flash point	-43°C	38°C
Speed	150 m/s	3m/s

- Explosion.
- properties of fuel (H_2 , ~~CH~~ Methane, propane)

→ Image processing of Hydrogen safety

→ Dust explosion.

→ Drawback of Hydrogen, Lithium (low energy density)

→ we want to use metal powders as fuel on earth & space. → (high energy densities)

→ Metals as zero-carbon energy carriers.

→ Microgravity experiments on drop towers. parabolic flights.

→ Fe particles 3 65% loss.

→ flame cells show instability.

Session 4 → Traditional Toys

Session 5 AI / CV {computer vision} / vision, Medical Imaging
→ Bissier Raytchev

→ Computer graphics; Image processing.

→ Computer Vision

→ Bag of features / framework

→ Colorectal Tumors Classification

→ Automated ~~IPs~~ IPS self clustering.

→ Vision in the ~~brain~~ brain.

→ CV & Deep vs flat processing

→ convolutional Neural networks

→ CNN explainer (playground)

→ vision & Language [TRANSFORMERS]

→ Deliver sight to the blind.